Sameeksha Sankpal

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EDUCATION

Vivekanand Education Society's Institute of Technology **August 2023 - Present**
Bachelor of Engineering in Information Technology (Honors in Research) | CGPA : 9.87 / 10.00 Mumbai, India
Honors Research (ongoing): Adversarial Noise Integration in Hybrid Quantum GANs for Synthetic Data Privacy

PUBLICATIONS & PREPRINTS

Sameeksha S. Sankpal, Chandresh K. Maurya, and Rahul K. Dubey. "Beyond Naturalness: Diagnosing the Synthetic-to-Real Gap in Spoken Language Understanding" *Under review, Transactions on Machine Learning Research, 2026.*

Sameeksha S. Sankpal, Shifa Shaikh, Shantanu Pawar, Anushka Sharma, Mannat Doultani, and Shardul Gore "Does Quantum Attention Help Forecast Stock Volatility? A Matched Quantum-Classical Study" *Under review, International Conference on Intelligent Networks and Data-driven Intelligent Applications.*

Sameeksha S. Sankpal, and Anushree Prabhu "The Generator or the Audit: Evaluation Design in Membership Inference on Synthetic Tabular Data" *Under review, 5th IEEE Conference on Secure and Trustworthy Machine Learning.*

Vishal Sharma, Chandresh K. Maurya, and Sameeksha S. Sankpal "What Actually Improves Low-Resource ASR: Lessons from Thirty-Seven LoRA Adaptations Across Eight Border Languages" *Under review, 22nd International Conference on Natural Language Processing.*

EXPERIENCE

Research Intern, Indian Institute of Technology Indore **December 2025 - May 2026**
- Adapted a text-and-speech synthesis pipeline to three SLU benchmarks across nine languages, then measured what synthetic-only training actually transfers to real test audio rather than reporting same-distribution cross-validation.
- Located the synthetic-to-real speech gap in speaker identity, not channel. Telephony channel-matching moved synth-only intent accuracy +0.9pp, while F5-TTS voice cloning moved it +22.3pp (49.6% → 76.2%). A confound-controlled diagnostic put the synth-specific identity penalty at 24.7pp against 1.3pp for channel.
- Isolated the text-side gap to pragmatic register through a five-seed ablation. First-person and causal phrasings gained +4.6 and +4.4pp intent accuracy, while politeness templates were null.
- Traced a Hindi/Turkish slot-tagging failure to BIO tokenization splitting on the Devanagari virama. The fix raised mBERT slot F1 by roughly 40pp on both low-resource languages.

Research Fellow, Amity University **June 2025 - July 2025**
- Built an affective-state recognition pipeline from physiological signals, using MRMR ranking with SHAP-based iterative pruning for feature selection before benchmarking classical and deep models.

Linux Kernel Bug Fixing Mentee, The Linux Foundation **June 2025 - November 2025**
- Selected for the LFX kernel mentorship. Authored C patches to the I2C, DVB media and seccomp subsystems and carried them through maintainer review on LKML.

PROJECTS

Qapital: Quantum Portfolio Selection on Real Market Data | *Link* **October 2026**
- Built and deployed a QAOA portfolio optimiser for NIFTY 50 and S&P 100 stocks with penalty, XY-mixer and warm-start circuits, scored against brute force, simulated annealing, a convex relaxation and random guessing..
- Across 64 quarterly problems since 2019, CVaR tuning beat expectation tuning on 58. Two warm-started layers lifted quality from 4.5× to 21.1× random guessing and improved on their classical start in all 64.
- Under an IBM noise model the XY circuit's edge fell from 17× to 1.5×. Mitigation and post-selection recovered 5×.

QuantFormer: Quantum Attention for Stock Volatility Forecasting (Capstone Project)
- Built a hybrid transformer with 12 variational circuits in its attention, plus a differentiable density-matrix simulator in PyTorch that matched Qiskit to 10^{-8} and replaced parameter-shift gradients with autodiff.
- Tuning the circuit learning rate raised out-of-sample R^2 on five-day volatility from 0.285 to 0.327 and halved seed variation, matching a classical twin and an exact 81-feature classical surrogate derived by us.

TECHNICAL SKILLS

- **Quantum**: PennyLane, Qiskit, Stim, Sinter, PyMatching, ldpc; density-matrix simulation, surface-code decoding
- **ML & AI**: PyTorch, Stable-Baselines3, scikit-learn, Transformers, NumPy, Pandas
- **Languages/Frameworks**: Python, C/C++, SQL, JavaScript, LaTeX

AWARDS, SERVICE AND LEADERSHIP

Runner Up, Manthan: Deep Thinking **March 2026**
- Technical paper presentation competition organizeed by VESIT IT Department; 5 finalists from 20 teams

Junior Technical Officer | QuestIT Cell **August 2024 - July 2025**
- Delivered a lecture on JavaScript runtime architecture to 100+ students and ran debugging support at workshops.